

Chapter 4 — Exercises

Multiple-Choice Questions

Representing data & engineering features

How to Use These Slides

- **12 multiple-choice** questions on Chapter 4 concepts
- Pick **one** answer (A–D) for each question
- The **answer key** with brief explanations is at the end
- Try to answer before revealing the solutions

Topics: categorical encoding, binning, interactions & polynomials, transformations, feature selection, and expert knowledge.

Q1. Why not encode categories as integers 0, 1, 2, ...?

- A. It implies a false order and spacing between categories
- B. Integers take too much memory
- C. scikit-learn rejects integer columns
- D. It always causes overfitting

Q2. One-hot encoding represents a categorical feature as:

- A. a single integer column
- B. one 0/1 column per category
- C. the mean of the target per category
- D. a floating-point average

Q3. By default, `pd.get_dummies` encodes:

- A. all numeric columns
- B. only the target column
- C. object / string (categorical) columns
- D. nothing until the data is scaled

Q4. Binning a continuous feature mainly benefits:

- A. linear models (it adds flexibility)
- B. decision trees
- C. random forests
- D. all models equally

Q5. An interaction feature is typically:

- A. the logarithm of a feature
- B. a feature minus its mean
- C. a one-hot column
- D. the product of two features

Q6. A log transform is especially helpful for:

- A. tree-based models
- B. skewed count data used with linear models
- C. already-Gaussian features
- D. categorical features

Q7. Univariate feature selection can miss features that are:

- A. strongly correlated with the target
- B. high-variance
- C. useful only in combination with other features
- D. numeric

Q8. Model-based selection (`SelectFromModel`) uses:

- A. a model's feature importances or coefficients
- B. a single statistical test per feature
- C. random sampling of features
- D. the mean of the target

Q9. Recursive Feature Elimination (RFE):

- A. is the cheapest selection method
- B. selects features without any model
- C. only works with linear models
- D. iteratively fits and drops the weakest feature

Q10. In the Citi Bike case, a random forest using only POSIX time:

- A. achieved near-perfect accuracy
- B. failed because trees can't extrapolate beyond the training range
- C. needed no features at all
- D. worked only after scaling

Q11. Which features dramatically improved the Citi Bike model?

- A. The raw timestamp squared
- B. Random noise features
- C. Hour of day and day of week
- D. The POSIX time in milliseconds

Q12. For a `LinearRegression` on hour/day, the best encoding was:

- A. one-hot encoding (plus interactions)
- B. raw integers
- C. no encoding at all
- D. standardized integers

Answer Key

Solutions & Explanations

Answers 1–6

Q	Ans	Why
1	A	Integer codes imply a false order the model would use
2	B	One-hot = one 0/1 column per category
3	C	<code>get_dummies</code> encodes string/object columns by default
4	A	Binning lets linear models fit nonlinear effects
5	D	An interaction feature is a product of features
6	B	Log tames skewed count data for linear models

Answers 7–12

Q	Ans	Why
7	C	Univariate tests miss features useful in combination
8	A	<code>SelectFromModel</code> uses importances / coefficients
9	D	RFE iteratively drops the weakest feature (expensive)
10	B	Trees can't extrapolate → POSIX time fails on new dates
11	C	Hour of day and day of week captured the pattern
12	A	One-hot (+ interactions) matched the random forest

Well done!

Review Chapter 4 for any you missed